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COURSE CODE: SBT-DF203-BASIC NETWORKING SKILLS

LAB NAME: SBT-DF203_Lab3_SYN_FLOOD_ATTACK_INVESTI

DATE: 11/09/26

INSTRUCTOR: AMINU IDRIS

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 3: SYN Flood Attack Investigation Using tshark

Three-Week Delivery Block 1/3 | 5-11 September 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.	
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 3
Lab Title	SYN Flood Attack Investigation Using tshark
Delivery Block	1/3 of 3
Scheduled Dates	5-11 September 2026
Estimated Duration	3-4 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	mySYNFloodCapture.pcap or a bounded loopback capture generated in the lab

1. Source Materials and Slide Alignment

- HTTP, tshark and SYN Flood: TCP three-way handshake, half-open connections, SYN flood concept, Scapy packet generation and tshark capture.
- The uploaded slide uses Apache on port 80, loopback capture and a four-packet controlled simulation.
- The assignment requires two authorized VMs or an equivalent isolated environment and evidence that distinguishes attack-like SYN activity from a normal handshake.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

An Apache service experienced repeated TCP SYN packets that were not followed by the final ACK of the three-way handshake. You will preserve the packet capture, establish a normal-traffic baseline, identify suspicious SYN patterns, calculate indicators, and conduct a strictly bounded loopback simulation to validate your interpretation.

3. Learning Outcomes

- Differentiate a complete TCP handshake from an incomplete or half-open attempt.
- Use tshark display filters and field extraction to identify SYN, SYN-ACK, ACK and reset packets.
- Calculate SYN counts, unique source ports, response behaviour and incomplete-handshake indicators.
- Create a four-packet loopback-only simulation with Scapy in an authorized environment.
- Develop a defensible timeline and incident finding from PCAP evidence.
- Recommend controls for detecting and mitigating SYN flood activity.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- The simulation is limited to four SYN packets against 127.0.0.1 or an instructor-assigned private lab IP. Do not increase packet count, automate continuous sending, spoof third-party addresses or target public systems.

5. Required Tools and Environment

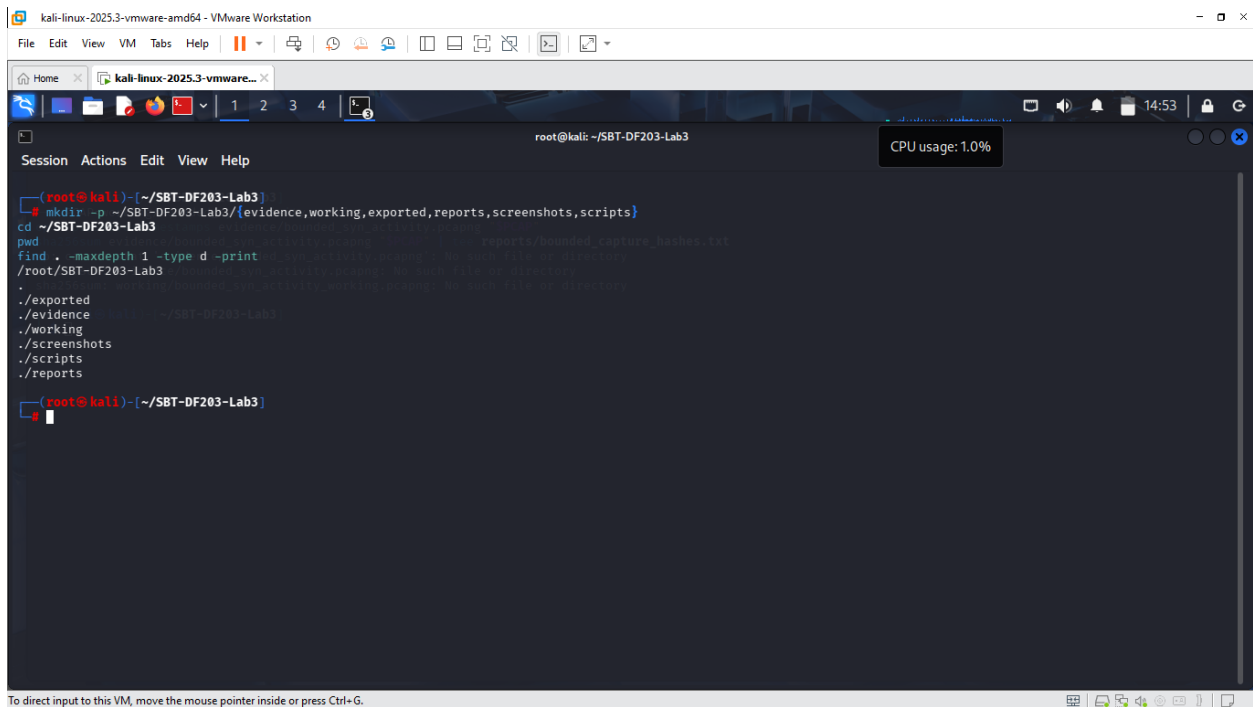
Item	Recommended Tool	Purpose
Operating system	Kali Linux VM	Run tshark, Apache and Scapy.
Target service	Local Apache on 127.0.0.1:80	Safe target for bounded simulation.
Capture tool	tshark/Wireshark	Capture and analyze TCP flags.
Packet library	python3-scapy	Generate a small number of SYN packets.
Evidence	Provided PCAP or student capture	Forensic analysis source.
Optional second VM	Authorized host-only target	Demonstrate network capture without touching external networks.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab3/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab3
pwd
find . -maxdepth 1 -type d -print
```

ANSWER:

ROOT@KALI



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab3 CPU usage: 1.0%
Session Actions Edit View Help
(root@kali)~[~/SBT-DF203-Lab3]
$ mkdir -p ~/SBT-DF203-Lab3/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab3
pwd
/root/SBT-DF203-Lab3
find . -maxdepth 1 -type d -print
./exported
./evidence
./working
./screenshots
./scripts
./reports
(root@kali)~[~/SBT-DF203-Lab3]
$
```

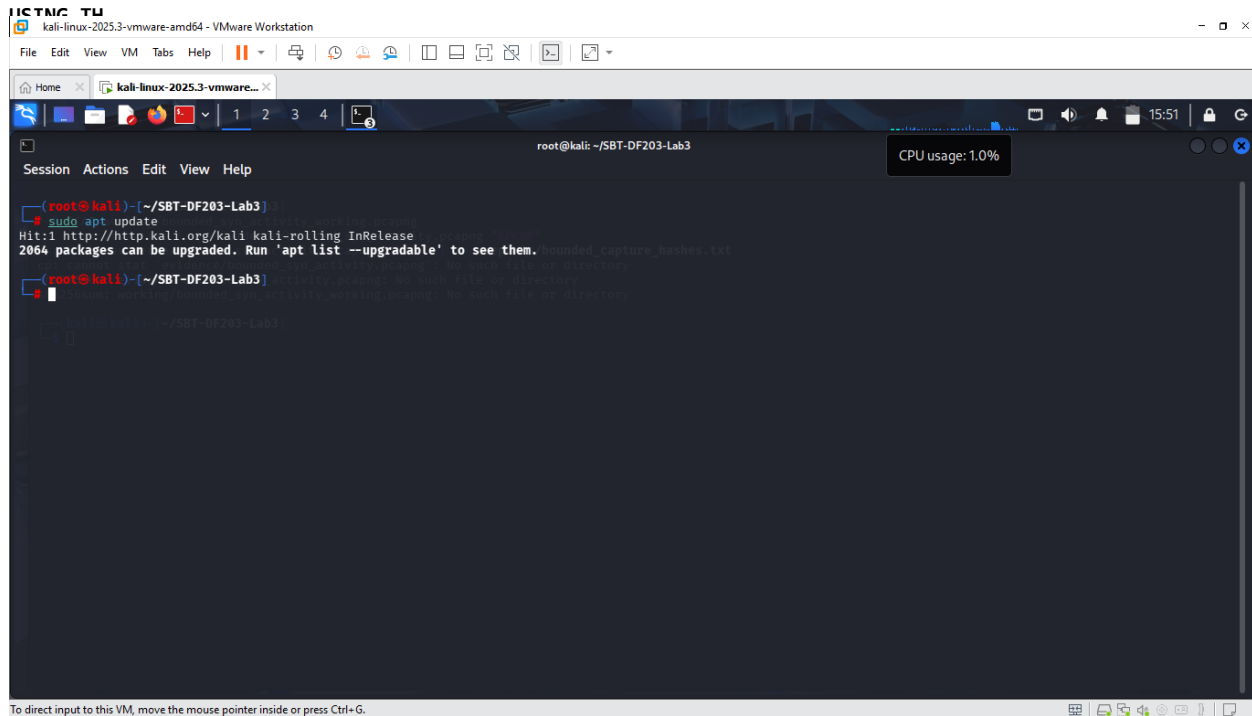
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

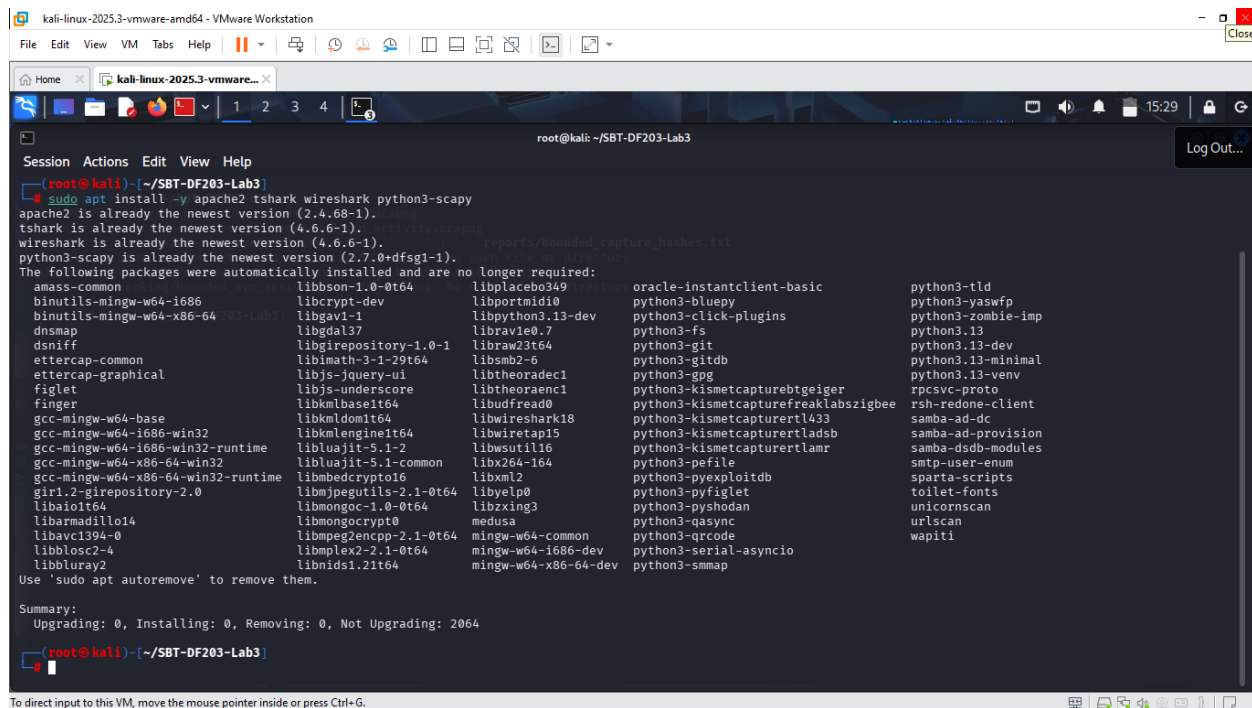
```
sudo apt update
sudo apt install -y apache2 tshark wireshark python3-scapy
sudo systemctl enable --now apache2
# Optional supplied capture
wget -O evidence/mySYNFloodCapture.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap'
sha256sum evidence/mySYNFloodCapture.pcap | tee reports/syn_capture_sha256.txt
```

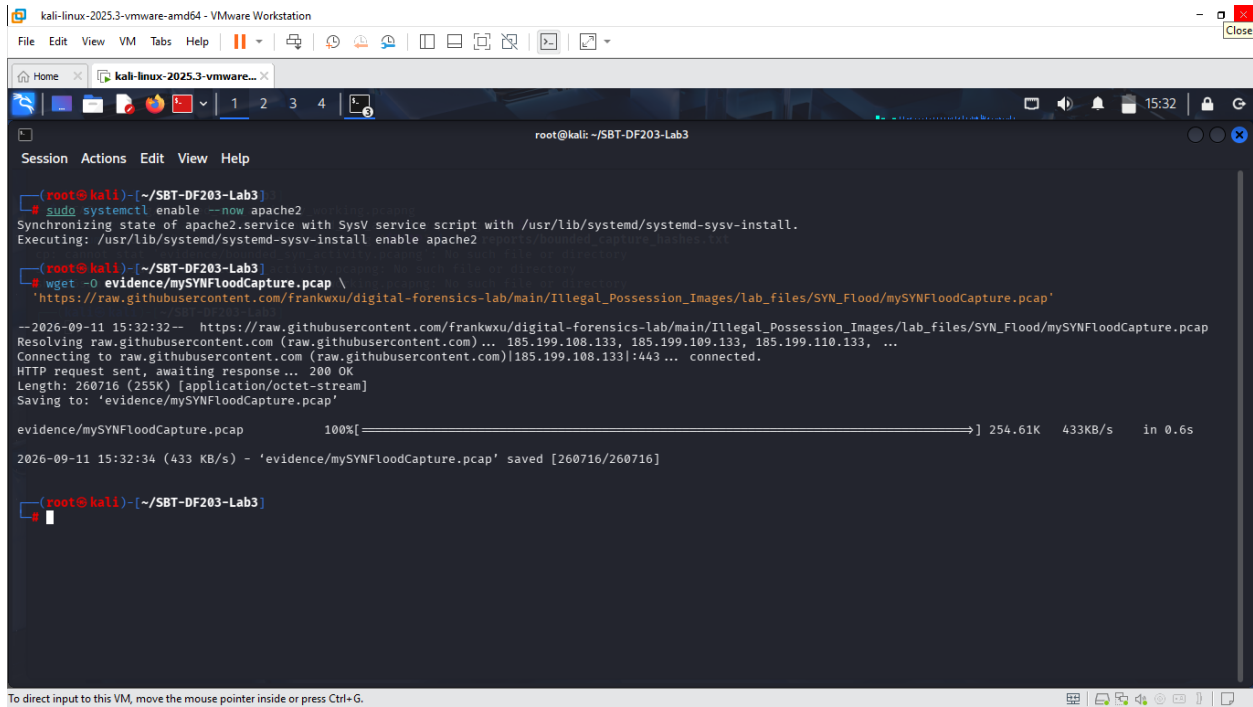
Screenshot required: The Apache service state, evidence file details and SHA-256 hash.

ANSWER:



E COMMAND ABOVE:





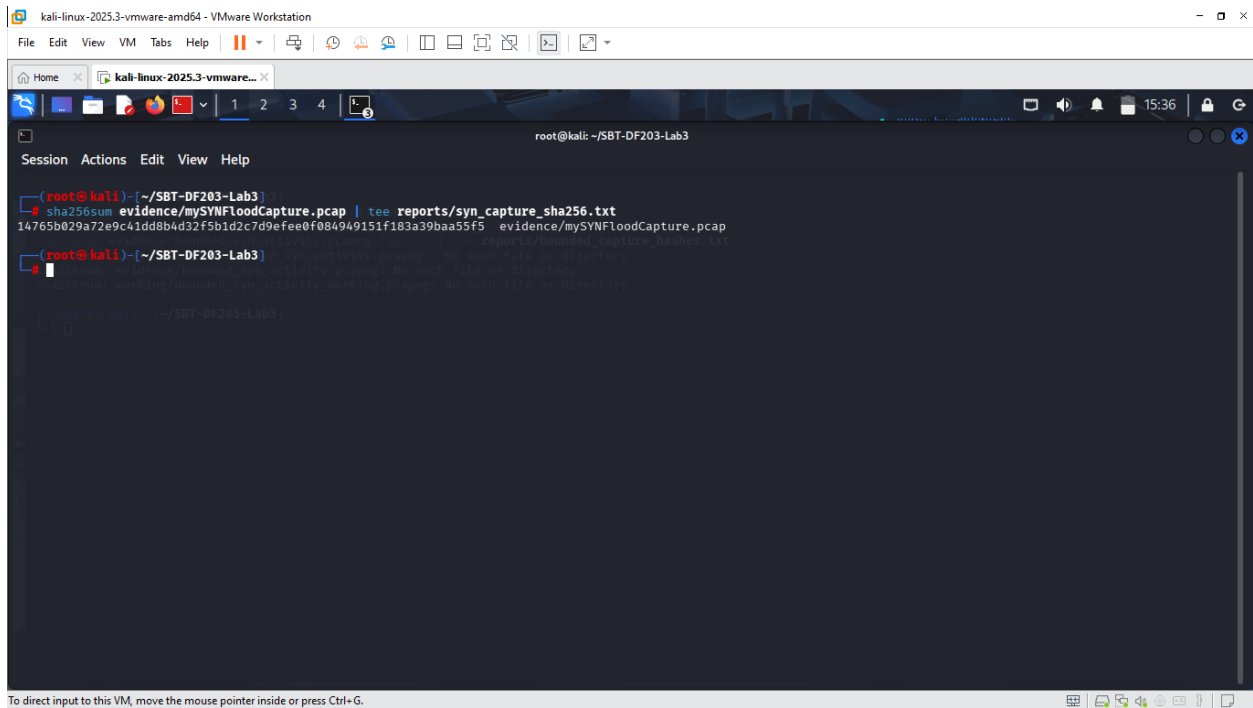
```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab3
Session Actions Edit View Help
(root@kali) ~/SBT-DF203-Lab3
# sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
(root@kali) ~/SBT-DF203-Lab3
# wget -O evidence/mySYNFloodCapture.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap'
--2026-09-11 15:32:32-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com) ... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 260716 (255K) [application/octet-stream]
Saving to: 'evidence/mySYNFloodCapture.pcap'

evidence/mySYNFloodCapture.pcap 100%[=====] 254.61K 433KB/s in 0.6s

2026-09-11 15:32:34 (433 KB/s) - 'evidence/mySYNFloodCapture.pcap' saved [260716/260716]

(root@kali) ~/SBT-DF203-Lab3
#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab3
Session Actions Edit View Help
(root@kali) ~/SBT-DF203-Lab3
# sha256sum evidence/mySYNFloodCapture.pcap | tee reports/syn_capture_sha256.txt
14765b029a72e9c41dd8b4d32f5b1d2c7d9efee0f084949151f183a39baa55f5 evidence/mySYNFloodCapture.pcap
(root@kali) ~/SBT-DF203-Lab3
#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab3-ADOYI NATHANIEL JOHNSON
Trainee name	ADOYI NATHANIEL JOHNSON
Date and time started	11/09/26
Evidence file name(s)	SBT-DF203_Lab3_SYN_FLOOD_ATTACK_INVESTIGATION USING TSHARK
Source or generation method	ROOT@KALI LINUX COMMAND LINE
Original SHA-256	sha256sum evidence/mySYNFloodCapture.pcap tee reports/syn_capture_sha256.txt
Working-copy SHA-256	sha256sum evidence/mySYNFloodCapture.pcap tee reports/syn_capture_sha256.txt 14765b029a72e9c41dd8b4d32f5b1d2c7d9efee0f084949151f183a39baa55f5 evidence/mySYNFloodCapture.pcap
Analysis workstation/VM	ROOT@KALI
Notes on any changes	NONE

8. Part A - Establish a Normal Handshake Baseline

```

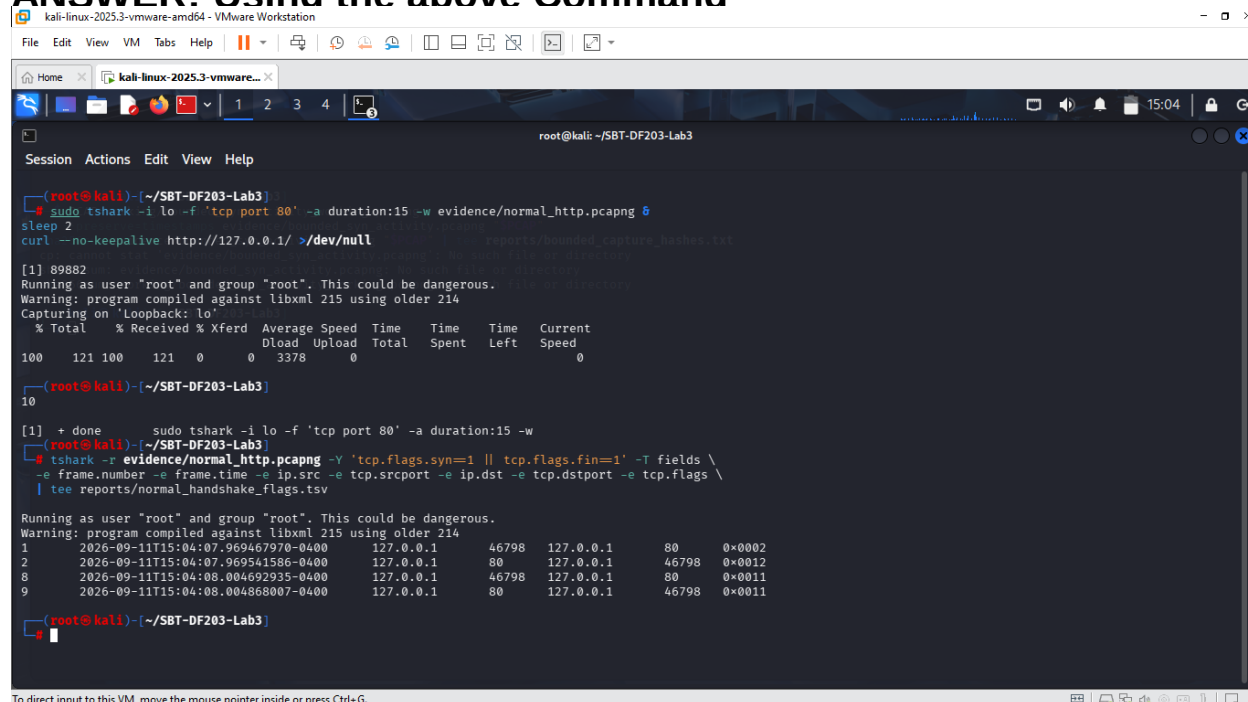
sudo tshark -i lo -f 'tcp port 80' -a duration:15 -w evidence/normal_http.pcapng &
sleep 2
curl --no-keepalive http://127.0.0.1/ >/dev/null
wait

tshark -r evidence/normal_http.pcapng -Y 'tcp.flags.syn==1 || tcp.flags.fin==1' -T fields \
  -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags \
  | tee reports/normal_handshake_flags.tsv

```

Document the expected SYN, SYN-ACK and ACK sequence before examining suspicious traffic.

ANSWER: Using the above Command



```
root@kali: ~/SBT-DF203-Lab3

[1] 89882
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed

100    121    100    121     0     0   3378      0    0      0    0     0      0

root@kali)~# sudo tshark -i lo -f 'tcp port 80' -a duration:15 -w evidence/normal_http.pcapng &
sleep 2
curl --no-keepalive http://127.0.0.1/ >/dev/null

root@kali)~# tshark -r evidence/normal_http.pcapng -Y 'tcp.flags.syn==1 || tcp.flags.fin==1' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags \
| tee reports/normal_handshake_flags.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1  2026-09-11T15:04:107.969467970-0400    127.0.0.1      80      0x0002
2  2026-09-11T15:04:107.969541586-0400    127.0.0.1      80      127.0.0.1    46798  0x0012
8  2026-09-11T15:04:108.004692935-0400    127.0.0.1      80      127.0.0.1    46798  0x0011
9  2026-09-11T15:04:108.004868007-0400    127.0.0.1      80      127.0.0.1    46798  0x0011
```

9. Part B - Conduct the Bounded Loopback Simulation

Create the following script exactly. It sends four SYN packets to the local Apache service and then exits. It does not run continuously and does not target another organization.

```
cat > scripts/syn_probe_lab.py <<'PYLAB'
from scapy.all import IP, TCP, RandShort, send

TARGET = '127.0.0.1'
PORT = 80
COUNT = 4

packets = [IP(dst=TARGET)/TCP(sport=RandShort(), dport=PORT, flags='S') for _ in range(COUNT)]
send(packets, verbose=False)
print(f'Sent {COUNT} authorized training SYN packets to {TARGET}:{PORT}')
PYLAB

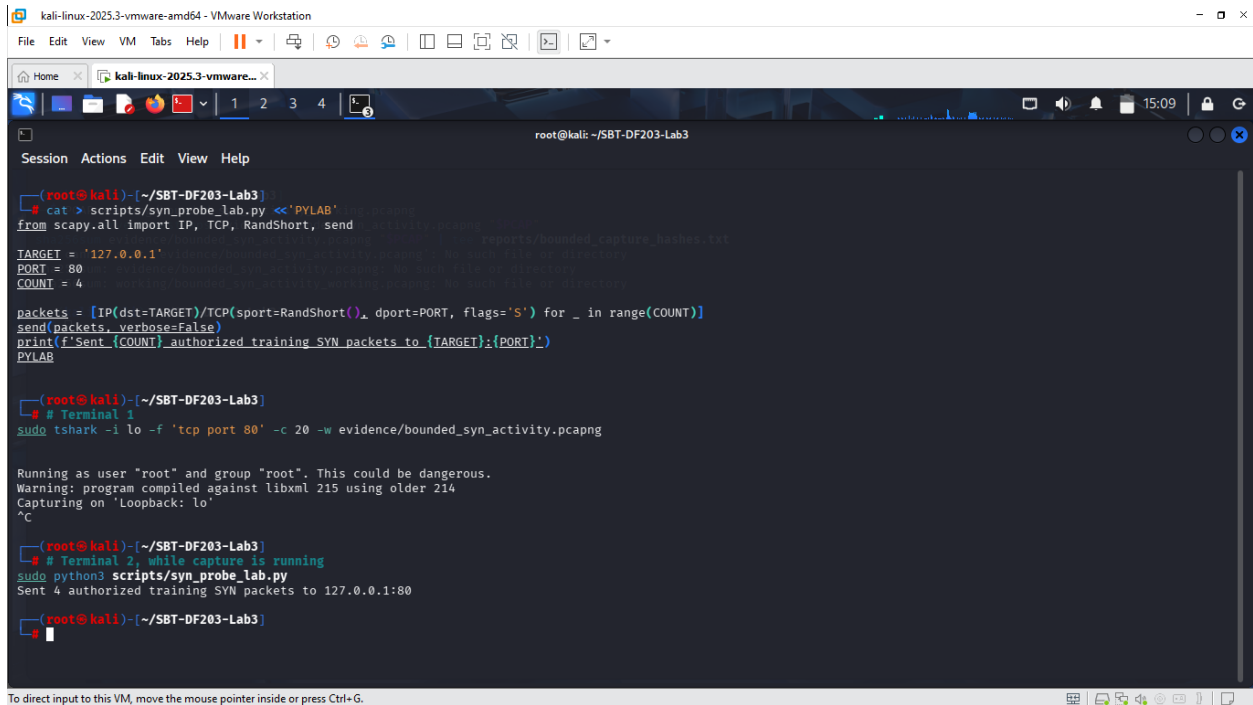
# Terminal 1
sudo tshark -i lo -f 'tcp port 80' -c 20 -w evidence/bounded_syn_activity.pcapng

# Terminal 2, while capture is running
sudo python3 scripts/syn_probe_lab.py
```

Stop Condition

If the capture reaches 20 packets it stops automatically. Otherwise stop with Ctrl+C immediately after the four SYN packets and associated responses are captured.

ANSWER:



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab3

Session Actions Edit View Help

(root@kali)~/~/SBT-DF203-Lab3
# cat > scripts/syn_probe_lab.py <<'PYLAB'
from scapy.all import IP, TCP, RandShort, send
reports/bounded_capture_hashes.txt
TARGET = '127.0.0.1'
PORT = 80
COUNT = 4
packets = [IP(dst=TARGET)/TCP(sport=RandShort(), dport=PORT, flags='S') for _ in range(COUNT)]
send(packets, verbose=False)
print(f'Sent {COUNT} authorized training SYN packets to {TARGET}:{PORT}')
PYLAB

(root@kali)~/~/SBT-DF203-Lab3
# Terminal 1
sudo tshark -i lo -f 'tcp port 80' -c 20 -w evidence/bounded_syn_activity.pcapng

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'
^C

(root@kali)~/~/SBT-DF203-Lab3
# Terminal 2, while capture is running
sudo python3 scripts/syn_probe_lab.py
Sent 4 authorized training SYN packets to 127.0.0.1:80

(root@kali)~/~/SBT-DF203-Lab3
```

10. Part C - Identify SYN Indicators with tshark

```
PCAP=working/bounded_syn_activity_working.pcapng
cp --preserve=timestamps evidence/bounded_syn_activity.pcapng "$PCAP"
sha256sum evidence/bounded_syn_activity.pcapng "$PCAP" | tee reports/bounded_capture_hashes.txt

# Initial SYNs only
tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \
  -e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.seq \
  | tee reports/initial_syns.tsv

# SYN-ACK responses
tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==1' -T fields \
  -e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.ack \
  | tee reports/syn_ack_responses.tsv

# Resets and final ACK candidates
tshark -r "$PCAP" -Y 'tcp.flags.reset==1 || (tcp.flags.ack==1 && tcp.len==0)' -T fields \
  -e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags \
  | tee reports/ack_reset_candidates.tsv
```



```
Session Actions Edit View Help
tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.seq \
| tee reports/initial_syns.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

(root@kali)~/SBT-DF203-Lab3
# SYN-ACK responses
tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==1' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.ack \
| tee reports/syn_ack_responses.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
tshark: "-Tfields" was specified, but no fields were specified with "-e".

(root@kali)~/SBT-DF203-Lab3
# Resets and final ACK candidates
tshark -r "$PCAP" -Y 'tcp.flags.reset==1 || (tcp.flags.ack==1 && tcp.len==0)' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags \
| tee reports/ack_reset_candidates.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

(root@kali)~/SBT-DF203-Lab3
#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

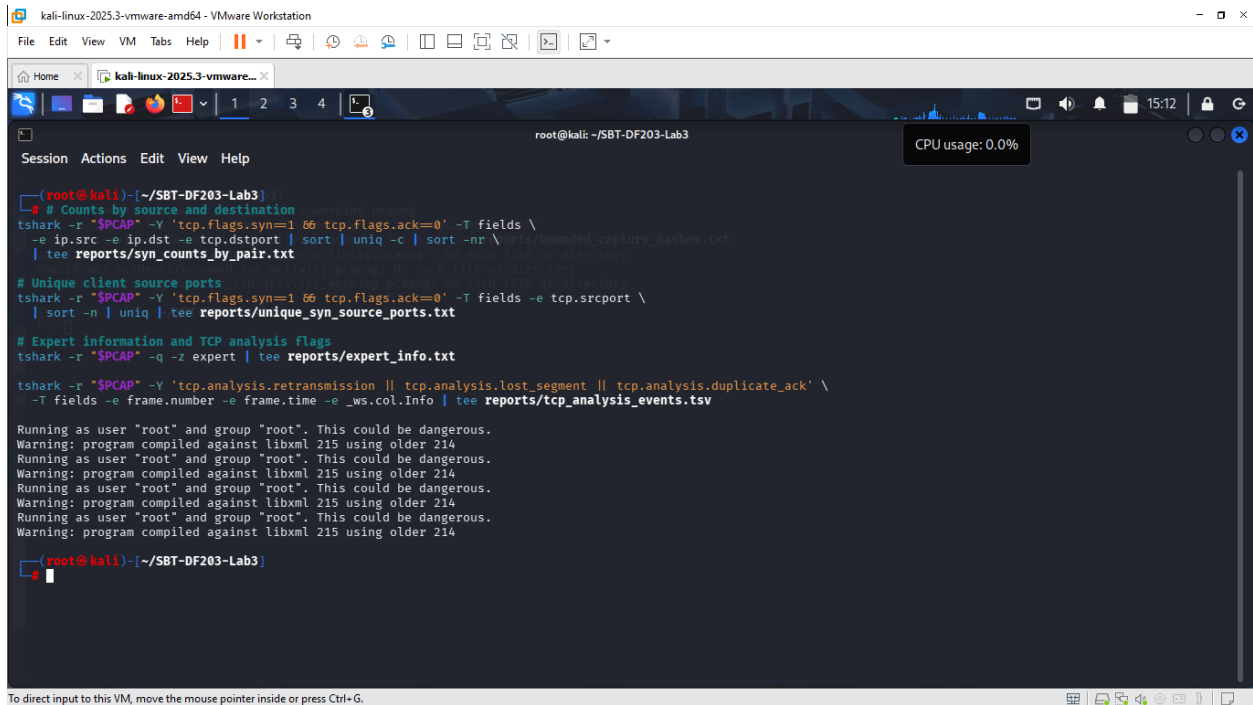
11. Part D - Quantify and Correlate the Activity

```
# Counts by source and destination
tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \
-e ip.src -e ip.dst -e tcp.dstport | sort | uniq -c | sort -nr \
| tee reports/syn_counts_by_pair.txt

# Unique client source ports
tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields -e tcp.srcport \
| sort -n | uniq | tee reports/unique_syn_source_ports.txt

# Expert information and TCP analysis flags
tshark -r "$PCAP" -q -z expert | tee reports/expert_info.txt

tshark -r "$PCAP" -Y 'tcp.analysis.retransmission || tcp.analysis.lost_segment ||
tcp.analysis.duplicate_ack' \
-T fields -e frame.number -e frame.time -e _ws.col.Info | tee reports/tcp_analysis_events.tsv
```



12. Part E - Compare Normal and Suspicious Sessions

Indicator	Normal HTTP Session	Bounded SYN Activity	Forensic Meaning
Initial SYN count			
SYN-ACK count			
Completed handshakes			
Unique client source ports			
HTTP request present?			
RST packets			
Observed duration			

A real SYN flood is defined by scale, rate, persistence and service impact. Four training packets are not a denial-of-service event; they reproduce the packet pattern in a safe way so that analysts can learn the indicators.

13. Detection and Mitigation Recommendations

- Alert on elevated SYN-to-completed-handshake ratios per source and destination.
- Monitor SYN backlog exhaustion, retransmissions, connection-state tables and service latency.
- Use SYN cookies, sensible backlog/time-out tuning, upstream rate limiting and DDoS protection where appropriate.
- Retain full packet capture or flow telemetry during incidents and synchronize system clocks.
- Correlate packet evidence with web server, firewall, load balancer and operating-system logs.

Required Screenshots Checklist

- ☐ Apache service and TCP port 80.
- ☐ Normal handshake in the baseline capture.
- ☐ Bounded Scapy script showing target and count.
- ☐ Initial SYN packet list.
- ☐ SYN-ACK response list.
- ☐ Any ACK or RST behaviour.
- ☐ Counts by source/destination.
- ☐ Unique source ports.
- ☐ Normal-versus-suspicious comparison table.
- ☐ Evidence hashes.

Student Report Template

1. Executive summary.
2. Authority, scope and safety controls.
3. Evidence description and hashes.
4. Normal handshake baseline.
5. Suspicious-traffic indicators and timeline.
6. Quantitative analysis.
7. Assessment of whether the capture proves a flood, scan or bounded simulation.
8. Detection and mitigation recommendations.
9. Appendix of commands and screenshots.

Submission Requirements

- One PDF report named SBT-DF203-Lab3_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Safety and evidence handling	15	Bounded loopback/host-only scope, hashes and clean stop.
Baseline analysis	15	Correct normal handshake identification.
tshark filtering	20	Accurate extraction of SYN, SYN-ACK, ACK/RST and fields.
Quantitative findings	20	Counts, ports, timing and correlation are correct.
Forensic interpretation	20	Clearly distinguishes packet pattern from proof of full DoS impact.
Report quality	10	Professional narrative and screenshots.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
Initial SYN filter	tcp.flags.syn==1 && tcp.flags.ack==0.
SYN-ACK filter	tcp.flags.syn==1 && tcp.flags.ack==1.
Normal handshake	Final ACK follows SYN-ACK and application data may follow.
Suspicious pattern	Many initial SYNs with few completed handshakes.

Checkpoint	Expected Observation
Training limitation	Four packets demonstrate indicators but do not prove service denial.
Strong conclusion	Requires rate, duration, system impact and corroborating logs.

Command Reference Summary

Command	Purpose
tshark -r file -Y FILTER	Analyze a saved capture.
tcp.flags.syn	Filter on the SYN flag.
tcp.flags.ack	Filter on the ACK flag.
tcp.stream	Conversation identifier.
tshark -q -z expert	Show expert analysis.
sort uniq -c	Count repeated field combinations.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.